PH101L Applied Physics Lab

Open-Ended Lab

Lab Report

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1. Introduction

The task involves analyzing and designing a circuit to meet a specific requirement for charging time;

- We are to verify that the capacitor charges up to 2.85 V (57% of 5 V) in 27.85 seconds.

2. Calculation

- All Calculations related to R and C by using charging/discharging equation of capacitors;

$$V(t) = Vmax\left(1 - e^{-t/RC}\right)$$

$$0.57 = 1 - e^{\frac{-t}{RC}}$$

$$0.57 + e^{\frac{-27.85}{RC}} = 1$$

$$e^{\frac{-27.85}{RC}} = 1 - 0.57 = 0.43$$

$$\ln e^{\frac{-27.85}{RC}} = \ln(0.43)$$

$$\frac{-27.85}{RC} = \ln(0.43)$$

$$RC = \frac{-27.85}{\ln(0.43)} = 33\,seconds$$

When using the formula for the charging of an RC circuit, we take the percentage to which the circuit is to be charged, in decimal form, and the **Vmax** is to be set at **1**.

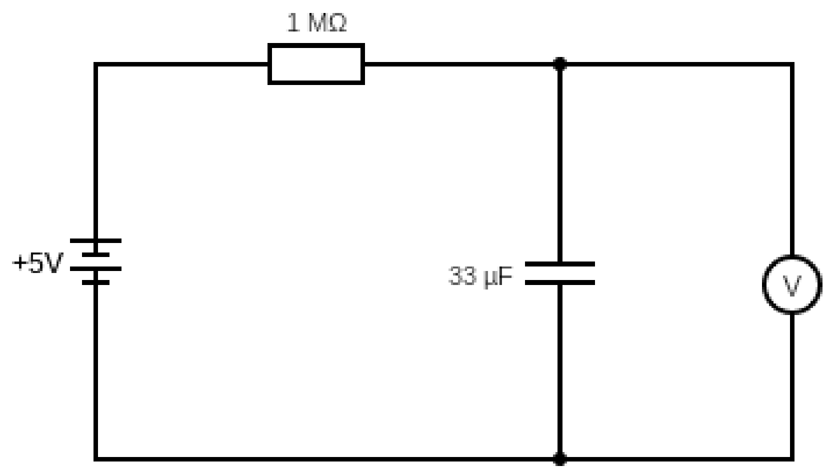
33 seconds = 33μF * 1MΩ

The **resistor** is to be taken of value **1 Mega ohm**, and the **capacitor** is to be taken of capacitance, **33 micro farads**.

[0.57 * 5 = 2.85 Volts]

3. Circuit Diagram

- Included below is the circuit diagram, which comprises of the 5 V power supply, the 33 micro farad capacitor, which is connected in parallel with the voltmeter, and lastly, the 1 mega ohm resistor.
- The circuit diagram is created and extracted from: [https://www.circuit-diagram.org/]



4. Data

A. Charging Voltage of a Capacitor

Time (t)	Voltage $V_C(t)$
5	0.68
10	1.29
15	1.77
20	2.17
25	2.54
30	2.82
35	3.07
40	3.27
45	3.45
50	3.58
55	3.71
60	3.81
65	3.89
70	3.96
75	4.02

B. Discharging Voltage of a Capacitor

Time (t)	Voltage $V_C(t)$
5	3.56
10	3.13
15	2.72
20	2.35
25	2.05
30	1.77
35	1.52
40	1.29
45	1.08
50	0.89
55	0.72
60	0.57
65	0.44
70	0.33
75	0.24

C. Charging Current of a Capacitor

Time (t)	Current $I_C(t)=C\frac{dv}{dt}$
5	4.488
10	4.026
15	3.168
20	2.64
25	2.442
30	1.848
35	1.65
40	1.32
45	1.188
50	0.858

55	0.66
60	0.528
65	0.462
70	0.396

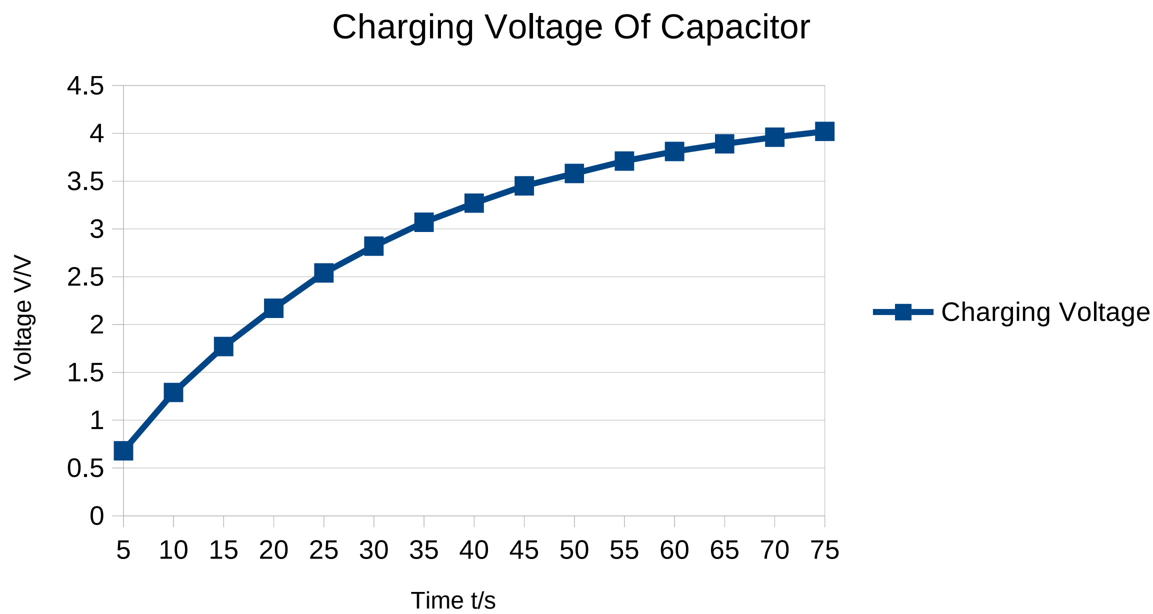
D. Discharging Current of a Capacitor

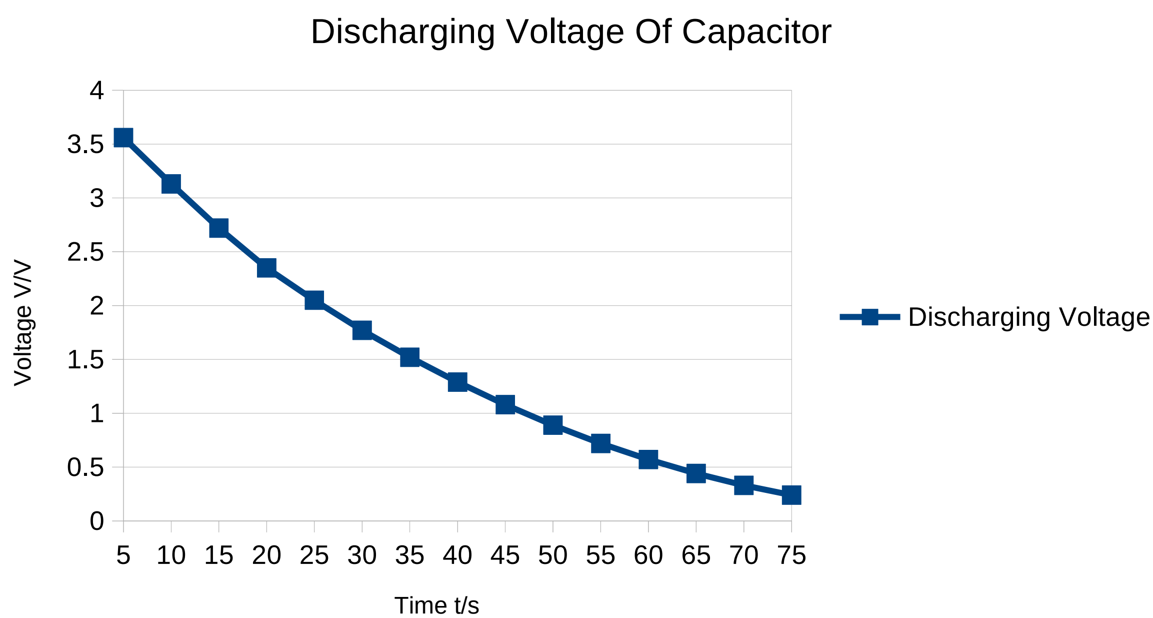
Time (t)	Current $I_C(t)=C\frac{dv}{dt}$
5	-3.036
10	-2.838
15	-2.706
20	-2.442
25	-1.98
30	-1.848
35	-1.65
40	-1.518
45	-1.386
50	-1.254
55	-1.122
60	-0.99
65	-0.858
70	-0.726
75	-0.594

5. Graphs

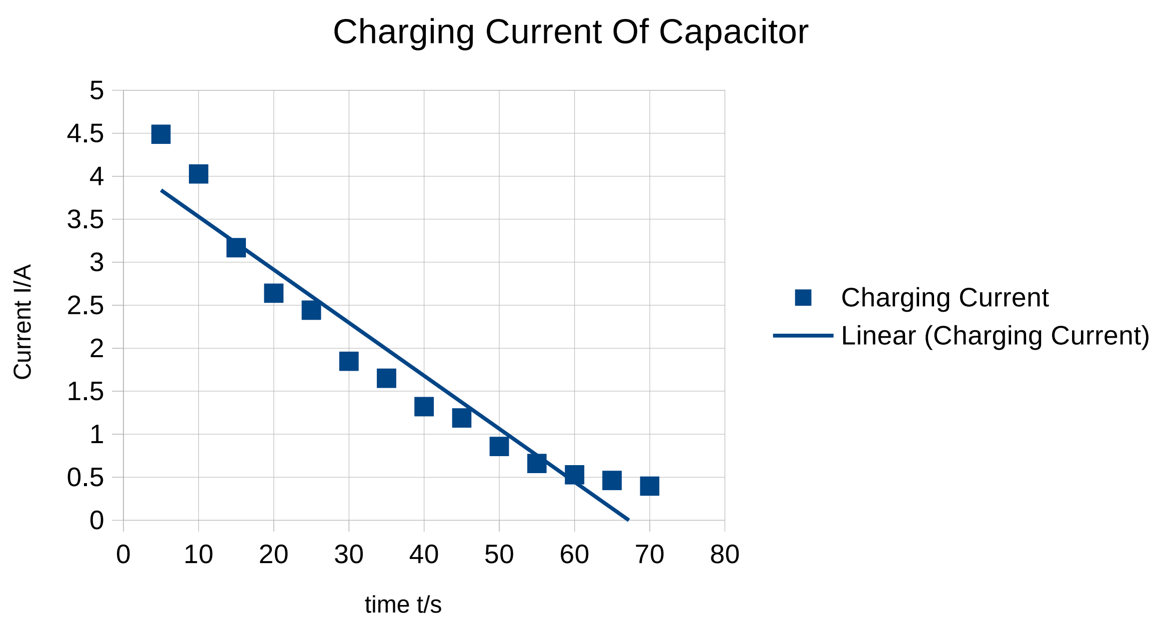
- Below displayed are the graphs of the respective requirements, plotted using LibreOffice Calc;

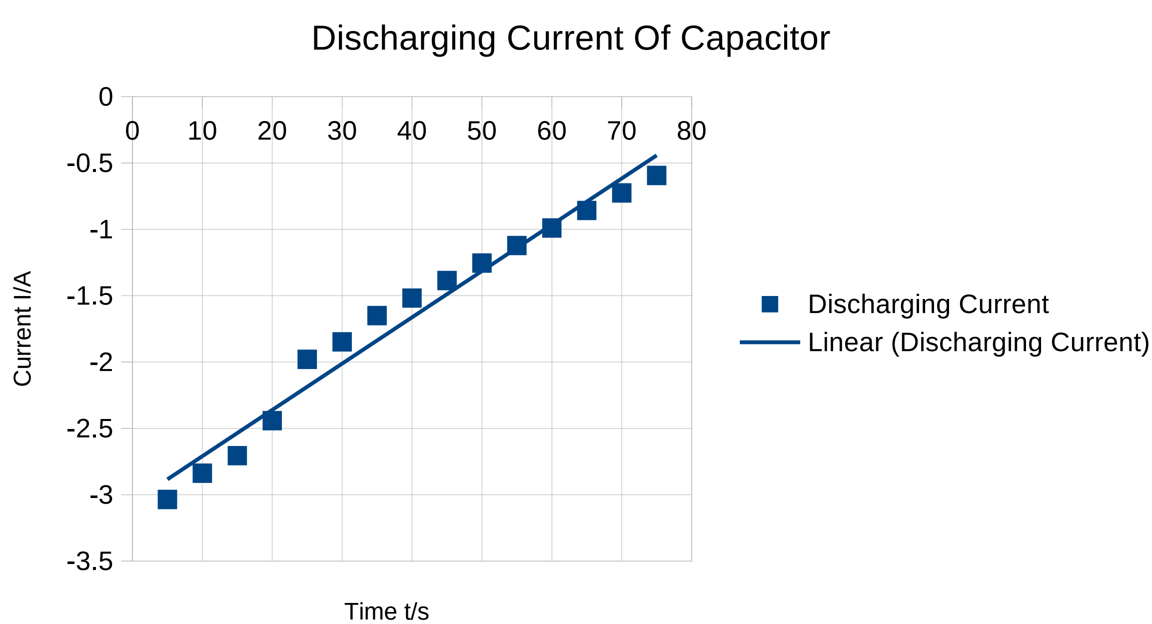
A. Charging and Discharging Voltage of a Capacitor


Charging Voltage Of Capacitor
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
Voltage V/V
5 10 15 20 25 30 35 40 45 50 55 60 65 70 75
Time t/s
Charging Voltage


Discharging Voltage Of Capacitor
4
3.5
3
2.5
2
1.5
1
0.5
0
Voltage V/V
5 10 15 20 25 30 35 40 45 50 55 60 65 70 75
Time t/s
Discharging Voltage

B. Charging and Discharging Current of a Capacitor





6. % Error

- Here, we compute the error pertaining to our experiment.

At 27.85 seconds, the capacitor voltage = 2.70 V (from graph)

57% of 5 V = 2.85 V (from the question statement)

$$\textbf{\textit{Percentage Error}} = \frac{Vmeasured - Vtrue}{Vtrue} \times 100$$

$$\textbf{\textit{Percentage Error}} = \frac{2.70 - 2.85}{2.85} \times 100 = 5.26\,\% \left(after\ taking\ the\ modulus \right)$$

$$\textbf{\textit{Percentage Error}} = \mathbf{5.26\%}$$

7. Conclusion

Since the percentage error is less than 10%, it is safe to say that it reflects a reasonable level of accuracy, and that the circuit charges up to 57% in 27.85 seconds.